AI-Driven Disaster Management and Dynamic Emergency Resource Allocation Using Machine Learning and GIS: A Flood-Response Framework for Chennai

Harsha Vardhini S
Department of Computer Science and Engineering
Rajalakshmi Engineering College
Chennai, India
harshavardhini.s.2024.cse@rajalakshmmi.edu.in

Harini Sankar
Department of Computer Science and Engineering
Rajalakshmi Engineering College
Chennai, India
harinisankar.2024.cse@rajalakshmmi.edu.in

Abstract—Urban floods overwhelm emergency authorities because high-risk areas are identified late, relief demand is estimated by hand, and supplies are often shared out by fixed plans rather than by actual need. This paper presents an AI-driven disaster management platform that integrates a GIS-based monitoring map, environmental indicators (rainfall, water level, flood severity, geographical vulnerability), machine-learning prediction of emergency resource demand, and a priority-weighted dynamic allocation engine, all surfaced through a centralised decision dashboard. Demand for food, drinking water, medical supplies, and relief materials is predicted per zone and distributed from depots according to predicted demand, severity, inventory, and accessibility instead of an equal split. The prototype covers four flood-prone zones in south Chennai: Tambaram East, Chromepet, Pallavaram, and Medavakkam. We formalise the allocation problem and evaluate the framework on a seeded synthetic scenario with assumed parameters. In that scenario, gradient-boosted and random-forest models predict demand with R^2 of 0.977 and 0.974 against 0.907 for linear regression, and when stock covers 60% of predicted demand the proposed rule delivers 72.9% fill to the most severe zone against 40.6% for an equal split, while using 100% of stock. We also report that it does not maximise the severity-weighted mean fill rate under scarcity, and discuss why.

Index Terms—disaster management, flood response, geographic information systems, machine learning, demand prediction, emergency logistics, resource allocation, decision support.

I. INTRODUCTION

Floods are among the most frequent and damaging natural hazards, affecting more people worldwide than almost any other disaster type [1]. Dense urban regions are especially exposed because of paved surfaces, constrained drainage, and large exposed populations. Effective response depends on three decisions made quickly under uncertainty: which areas are most affected, how much of each essential resource each area will need, and how limited stock should be moved from depots to those areas [2], [3]. International frameworks stress that early warning and prepositioned response capacity are central to reducing disaster losses [4].

In practice these decisions are frequently made with manual assessments, predefined distribution plans, and delayed field information. The result can be unequal distribution, with some areas over-supplied while others wait. Forecasting and monitoring tools exist, and machine learning has been applied to flood prediction [5], [6], but these capabilities are usually built separately from the logistics that follow. Monitoring says where water is rising; it does not say how many water litres, food packets, and medical kits each zone needs or from which depot they should be dispatched.

This paper proposes an integrated platform that closes this gap. The main contributions are:

- a GIS-based monitoring layer that visualises zones, roads, resource depots, and affected areas for location-aware planning;
- a transparent zone severity index built from rainfall intensity, water level, flood severity, and geographical vulnerability;
- a machine-learning model that predicts zone-wise demand for food, drinking water, medical supplies, and relief materials;
- a severity-priority dynamic allocation procedure that accounts for predicted demand, inventory, and accessibility and re-plans as conditions change; and
- a centralised dashboard and an evaluation that compares the allocation with equal-split, population-share, and max-min-fair baselines, including a candid account of where the proposed rule is not better.

Section II reviews related work; Section III formulates the problem; Section IV describes the method; Section V presents the architecture; Section VI describes implementation; Section VII reports experiments; Section VIII discusses the results; Section IX covers limitations; Section X concludes.

II. RELATED WORK

A. Flood Prediction and Machine Learning

Machine-learning approaches to flood prediction, including tree ensembles, neural networks, and hybrid models, have been surveyed extensively and generally outperform purely physical models when data are limited [5], [6]. Random forests [7] and gradient-boosted trees [8] are popular for tabular environmental data because they handle non-linear interactions and mixed feature scales with little tuning, and recurrent networks have shown strong rainfall–runoff skill when long gauge records exist [9]. Most of this literature predicts the hazard itself (water level, discharge, inundation extent), not the humanitarian demand that follows from it, which is the focus of this work.

B. Emergency Logistics and Relief Allocation

Operations research has long studied disaster logistics, covering facility location, prepositioning, and relief distribution [2], [3]. Models for multi-commodity relief distribution under urgent demand and limited transport capacity are well established [10], [11]. A recurring finding is that post-disaster logistics differs from commercial logistics: demand is uncertain, information arrives late, and the objective is to reduce human suffering rather than cost [12], [13]. Minimum standards for water, food, and health supplies provide reference levels for per-person demand [14]. Many of these models assume that demand is given; coupling them with a learned demand estimate is the direction taken here.

C. GIS-Based Decision Support

Geographic information systems let authorities overlay hazards, populations, roads, and facilities on one map, and crowd-sourced geographic data has become a practical complement to official sources [15], [16]. Road-network travel time between depots and zones is commonly computed with shortest-path algorithms [17], and this work uses the same idea to represent accessibility.

TABLE I
POSITIONING OF THE PROPOSED WORK

Capability	Flood ML [5],[6]	Relief OR [10],[11]	GIS tools [15],[16]	This work
Hazard monitoring	Yes	No	Partly	Yes
Demand prediction	No	Assumed given	No	Yes (ML)
Allocation by need	No	Yes	No	Yes
Re-plans on updates	Rarely	Rarely	Display only	Yes
Operator dashboard	Rarely	Rarely	Yes	Yes

D. Research Gap

Existing systems typically provide monitoring, or forecasting, or optimisation, but rarely the whole chain from environmental condition to predicted demand to adaptive distribution in a single operator-facing tool. This paper targets that chain at the scale of a city sub-region and evaluates it against simple, widely used allocation rules.

III. PROBLEM FORMULATION

A. System Model

Let Z be the set of monitored zones, D the set of resource depots, and $K = \{\text{food, water, medical, relief}\}$ the resource types. At decision time t, zone z has an environmental state vector $x_z(t)$, a population N_z, and a severity $S_z(t) \in [0,1]$. Depot j holds inventory $I_{jk}(t)$ of resource k. The travel time from depot j to zone z on the current road network is $c_{jz}(t)$, taken as infinite if the zone is unreachable. For a road graph $G = (V, E)$ with edge travel times $\tau_e(t)$, accessibility is the shortest-path time [17]

$$c_{jz}(t) = \min_{*P* \in paths(*j*,*z*)} \Sigma_{*e* \in *P*} \tau_e(t) \quad (1)$$

B. Allocation Objective

Let $\hat{d}_{zk}$ be the predicted demand and $x_{jzk} \geq 0$ the quantity of resource k sent from depot j to zone z. The unmet demand is

$$u_{zk} = \max(0, \hat{d}_{zk} - \Sigma_j x_{jzk}) \quad (2)$$

and the planner seeks to

$$\min \ \Sigma_z \Sigma_k w_z u_{zk} + \lambda \Sigma_{j,z,k} c_{jz} x_{jzk} \quad (3)$$

subject to the inventory constraint $\Sigma_z x_{jzk} \leq I_{jk}$ for every depot and resource, and $x_{jzk} = 0$ where c_{jz} is infinite. The weight w_z encodes zone priority (Section IV-D) and $\lambda \geq 0$ trades response cost against unmet need. This is a linear program and can be solved exactly; Section IV-D gives a closed-form rule that is easier to explain to officials.

C. Performance Measures

For a plan with allocations $a_{zk} = \Sigma_j x_{jzk}$, the fill rate of zone z and the severity-weighted fill rate are

$$\varphi_z = (1/|K|) \Sigma_k \min(a_{zk}, \hat{d}_{zk}) / \hat{d}_{zk} \quad (4)$$

$$\Phi = \Sigma_z S_z \varphi_z / \Sigma_z S_z \quad (5)$$

Stock utilisation is $U = \Sigma \min(a_{zk}, \hat{d}_{zk}) / \Sigma a_{zk}$, the share of dispatched stock that meets real demand. Inequality of coverage is measured with the Gini coefficient [20] of the fill rates,

$$G = (2 \Sigma_{i=1}^{n} i \varphi_{(i)}) / (n \Sigma_i \varphi_i) - (n+1)/n \quad (6)$$

where $\varphi_{(i)}$ are the fill rates sorted in ascending order and $n = |Z|$. We also report the Spearman rank correlation ρ between zone severity and fill rate: a plan that serves the worst-hit zones first has ρ near +1.

D. Requirements

The platform must be (1) *timely*, updating when conditions change; (2) *need-driven*, so that allocation follows severity and demand rather than a fixed share; (3) *explainable*, so that officials can see why a zone received a given quantity; and (4) *operable* on modest infrastructure through a single dashboard.

IV. PROPOSED METHOD

A. Monitoring Inputs and Normalisation

For each zone the system tracks rainfall intensity, water level, flood severity, weather conditions, and geographical vulnerability (for example low elevation and proximity to water bodies), together with population and past resource consumption. Each raw indicator r is min–max normalised to [0,1] using reference bounds:

$$\tilde{r} = (r - r_{min}) / (r_{max} - r_{min}) \quad (7)$$

B. Zone Severity Index

Four normalised indicators are combined into a transparent severity index:

$$S_z = \beta_1 R_z + \beta_2 W_z + \beta_3 F_z + \beta_4 V_z \quad (8)$$

where R_z is rainfall intensity, W_z water level, F_z flood severity, V_z geographical vulnerability, and $\Sigma \beta_i = 1$. The weights are policy parameters that can be tuned with local experts. In the experiments we use $\beta = (0.25, 0.25, 0.30, 0.20)$. To avoid reacting to sensor noise, severity is smoothed over time:

$$\tilde{S}_z(t) = \alpha\, S_z(t) + (1-\alpha)\, \tilde{S}_z(t-1) \quad (9)$$

with smoothing factor $\alpha \in (0,1]$; a re-plan is triggered when $|\tilde{S}_z(t) - \tilde{S}_z(t_{last})| > \theta$ for any zone.

C. Demand Prediction

A supervised regression model estimates demand for each resource from the zone state:

$$\hat{d}_{zk} = f_k(x_z, N_z, S_z, h_z) \quad (10)$$

where h_z summarises historical consumption in comparable events. Conceptually, demand factors into the affected population and a per-person norm:

$$d_{zk} = N_z \cdot \varphi(S_z) \cdot q_k \cdot h_z \cdot \varepsilon \quad (11)$$

where $\varphi(\cdot)$ is the share of the population affected as a function of severity, q_k is the per-person daily requirement for resource k (Table II), and ε is multiplicative noise. The learned models need not know this structure; they must recover it from data. Candidate learners are linear regression, random forests [7], and gradient-boosted trees [8], evaluated with normalised mean absolute error, normalised root-mean-square error, and R^2:

$$\text{nMAE} = (1/\bar{d}) \cdot (1/n) \Sigma\, |d_i - \hat{d}_i| \quad (12)$$

$$\text{nRMSE} = (1/\bar{d}) \cdot \sqrt{[(1/n) \Sigma\, (d_i - \hat{d}_i)^2]} \quad (13)$$

$$R^2 = 1 - \Sigma\, (d_i - \hat{d}_i)^2 \,/\, \Sigma\, (d_i - \bar{d})^2 \quad (14)$$

D. Severity-Priority Dynamic Allocation

Zone priority is the zone's share of total severity. Population is deliberately not included, because it already scales the predicted demand through Eq. (11):

$$w_z = S_z \,/\, \Sigma_y\, S_y \quad (15)$$

For each resource k with available stock I_k, the first-pass allocation to zone z is

$$a_{zk} = \min(\, \hat{d}_{zk}\,,\, I_k \cdot w_z\, \hat{d}_{zk} \,/\, \Sigma_y\, w_y\, \hat{d}_{yk}\,) \quad (16)$$

Any stock left after capping zones at their predicted demand is redistributed among zones with remaining unmet demand by repeating Eq. (16) until the stock is exhausted or all demand is met (a water-filling procedure). When stock is sufficient, every zone is filled completely. When stock is scarce, the fill rate of a zone is approximately proportional to its severity, so the worst-hit zones are covered best. Quantities a_{zk} are then assigned to depots by lowest travel time c_{jz}, subject to inventory. Because Eq. (16) is closed-form, every number shown to an official can be traced to severity and predicted demand.

E. Dynamic Re-Allocation

When new monitoring data arrive, severity, demand predictions, and road availability are refreshed and the allocation is recomputed. Algorithm 1 summarises the loop.

Algorithm 1. Dynamic Allocation Pipeline

```
INPUT zones Z, depots D, stock I, beta, alpha, theta
loop on each monitoring update:
  for each zone z: S_z <- Eq. 8; smooth by Eq. 9
  if no |dS| > theta: continue
  for each z, k: predict d_zk        # Eq. 10
  compute priority w_z               # Eq. 15
  a_zk <- water-filling of Eq. 16
  assign a_zk to depots by min c_jz  # Eq. 1
  publish plan to dashboard
end loop
```

V. SYSTEM ARCHITECTURE

The platform is organised in five layers (Fig. 1).

Data and monitoring layer: ingests rainfall, water level, weather, and population inputs and stores zone state.

- *GIS layer:* maintains the interactive map of the four zones, roads, depots, and affected areas, and computes depot-to-zone accessibility (Eq. 1).
- *Machine-learning layer:* trains and serves the demand-prediction model.
- *Allocation layer:* computes severity, priority, and the allocation plan and re-plans on updates.
- *Dashboard layer:* presents zone-wise disaster information, depot locations, predicted requirements, inventory, allocations, and emergency priorities to decision-makers.

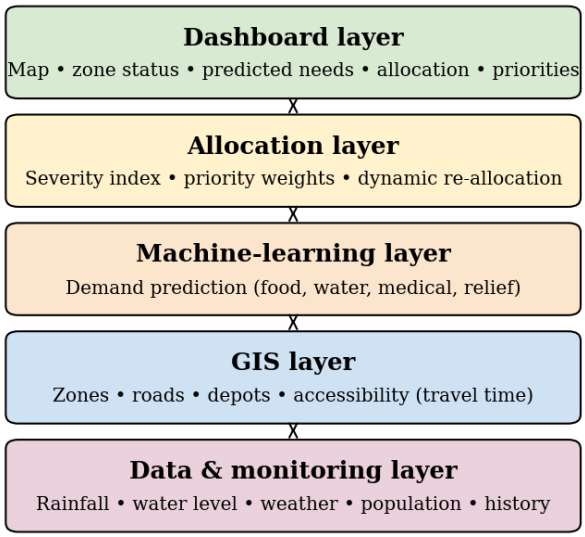


Fig. 1. Layered architecture of the proposed disaster management platform.

Fig. 2 shows the schematic spatial model used in the scenarios: four zones shaded by severity and three depots connected to every zone by candidate routes. The layout is illustrative and is not a map; the deployed GIS layer uses real coordinates and road geometry.

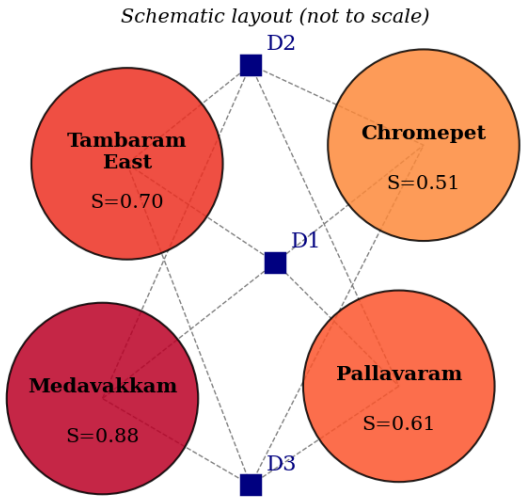


Fig. 2. Schematic of the four monitored zones (shaded by severity S for the severe scenario) and three depots D1–D3. Not to scale.

VI. IMPLEMENTATION

A. Prototype and Dashboard

The prototype monitors four zones in south Chennai: Tambaram East, Chromepet, Pallavaram, and Medavakkam, which include flood-prone localities. The dashboard shows an interactive map of zones, roads, depots, and affected areas; zone-wise disaster information; predicted requirements; available inventory; allocation results; and emergency priorities. Tech stack - Backend: Python (FastAPI) | Map Library: Leaflet.js | ML Library: Scikit-learn | Database: PostgreSQL

B. Machine-Learning Pipeline

Models are trained with scikit-learn [18]. Each resource has its own regressor. Inputs are rainfall, water level, flood severity, vulnerability, population, and a historical-use factor. Data are split 80/20 into training and held-out test sets with a fixed random seed for reproducibility.

C. Operating Procedure

During an event the monitoring layer pushes updates; the severity index is recomputed and smoothed (Eq. 9); if the change exceeds the trigger θ, demand is re-predicted and the allocation recomputed. Officials can inspect the severity, priority share, and predicted demand behind each recommendation, and can override a recommendation, with the override logged.

VII. EXPERIMENTS AND RESULTS

A. Experimental Setup

Because no public dataset links environmental conditions to relief consumption for these zones, we built a seeded synthetic dataset of 4,000 samples generated from Eq. (11) with a logistic affected-population function, per-person norms inspired by minimum humanitarian standards [14], and 8% multiplicative noise. Population and vulnerability values (Table II) are assumptions. Absolute numbers are therefore illustrative; the relative comparisons and sensitivities are the main findings.

TABLE II
ZONE PARAMETERS AND PER-PERSON NORMS (ASSUMED)

Zone	Population	Vulnerability V	Severity S (scenario)
Tambaram East	120,000	0.70	0.70
Chromepet	95,000	0.55	0.51
Pallavaram	110,000	0.60	0.61
Medavakkam	130,000	0.85	0.88

Resource	Food	Water	Medical / Relief
Per person per day	3 packets	15 litres	0.02 / 0.25 kits

B. Demand Prediction Accuracy

Table III and Fig. 3–4 report accuracy on the 800 held-out samples, averaged over the four resources. The tree ensembles reduce error by more than half relative to linear regression, which cannot represent the S-shaped relationship between severity and affected population. Gradient boosting is marginally better than the random forest. For water demand, the random forest assigns 74% of its importance to flood severity and 13% to rainfall, with the other four inputs about 3% each (Fig. 3b), consistent with the way the data were generated; on real data this ranking would need to be re-examined.

TABLE III
DEMAND-PREDICTION ACCURACY (HELD-OUT, SYNTHETIC)

Model	nMAE (%)	nRMSE (%)	R^2
Linear regression	20.3	25.7	0.907
Random forest	8.8	13.7	0.974
Gradient boosting	8.4	12.7	0.977

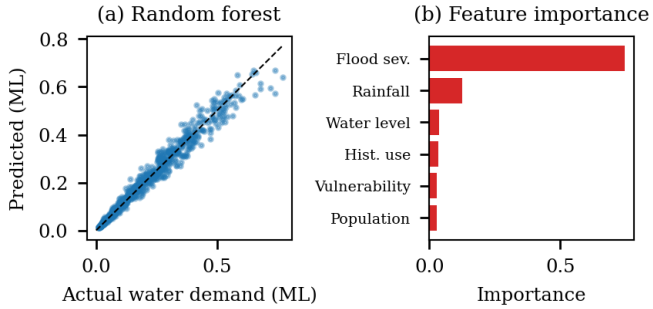


Fig. 3. (a) Predicted versus actual water demand (random forest, test set; axes in millions of litres). (b) Feature importance for water demand.

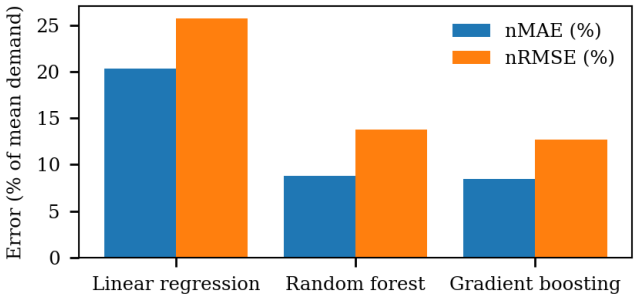


Fig. 4. Error of the three models relative to mean demand.

C. Allocation Under Scarcity

For the severe scenario (Table II, last column), the random-forest predictions give a total daily demand of about 353,000 food packets, 1.66 million litres of water, 2,262 medical kits, and 28,900 relief kits, with Medavakkam the largest and most severe. Stock is set to 60% of predicted demand for each resource. We compare five rules: *equal split* ($I_k/|Z|$ per zone, no redistribution of surplus); *population share*; *max-min fair* (equal weights with water-filling [19]); the proposed rule with priority S_z; and a variant with priority $S_z \cdot N_z$, which counts population twice because demand already scales with it. The variant is reported because it was our first formulation and it performed worse, so we replaced it in Eq. (15).

TABLE IV
ALLOCATION COMPARISON AT 60% STOCK (SYNTHETIC)

Rule	Φ (%)	Top-zone fill (%)	Min fill (%)	G	U (%)	ρ
Equal split	60.9	40.6	40.6	0.18	95.7	−1.0
Population share	63.1	46.4	46.4	0.16	99.8	−1.0
Max-min fair	60.0	60.0	60.0	0.00	100	n/a
Proposed (S·N)	57.4	78.7	33.2	0.17	100	1.0
Proposed (S)	58.3	72.9	42.1	0.11	100	1.0

Table IV and Fig. 5 show the trade-off honestly. The proposed rule ranks coverage in the same order as severity ($\rho = 1.0$), gives the most severe zone, Medavakkam, 72.9% fill against 40.6% under an equal split and 46.4% under population share, and uses all stock ($U = 100\%$, whereas the equal split strands 4.3% on Chromepet, whose demand is below its equal share). It also has a lower Gini coefficient than the equal and population rules. However, its severity-weighted mean fill Φ is 58.3%, slightly below the equal split (60.9%) and population share (63.1%). The reason is that Φ is not demand-weighted: equal and population rules fully serve the small-demand zone Chromepet (100%), which lifts the average, while the proposed rule deliberately gives Chromepet only 42.1% so that more reaches Medavakkam. Whether that trade is desirable is a policy decision; the platform makes it visible.

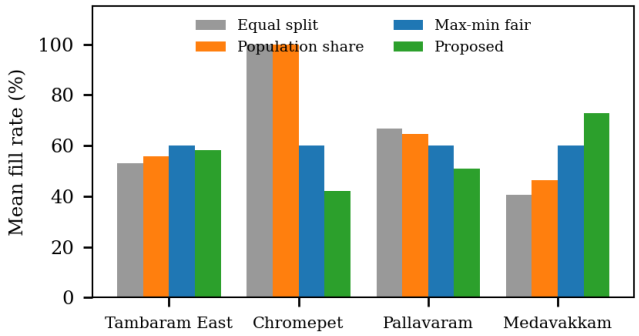


Fig. 5. Mean fill rate by zone for four allocation rules at 60% stock.

Fig. 6 shows the same decision for drinking water in absolute terms. Medavakkam needs about 0.59 million litres per day and Chromepet about 0.21 million. An equal split gives each zone a quarter of the stock, which is more than Chromepet needs and far less than Medavakkam needs; the proposed rule moves the difference toward the zones with the greatest severity-weighted demand.

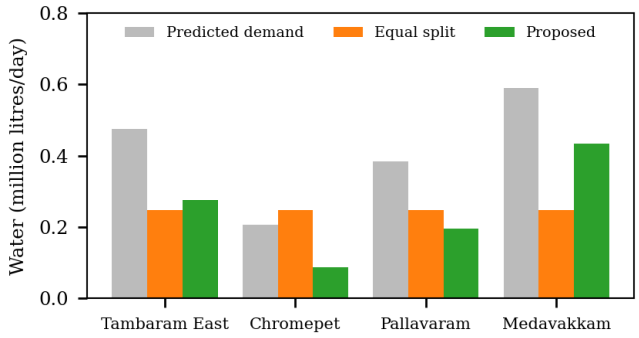


Fig. 6. Predicted water demand and allocated water per zone at 60% stock.

D. Sensitivity to Available Stock

Fig. 7 varies stock from 30% to 100% of predicted demand. At 30%, all rules have similar Φ (29–33%), and the proposed rule is lowest (29.1%) for the same reason as above. As stock grows the curves separate: at 60% the values are 60.9 (equal), 63.1 (population), 60.0 (max-min), and 58.3 (proposed), and at 100% the proposed and max-min rules reach 100% while equal split and population share reach only 86.4% and 90.8%, because their fixed shares leave some zones short and others over-supplied. The practical message is that need-based redistribution matters most when stock is close to demand.

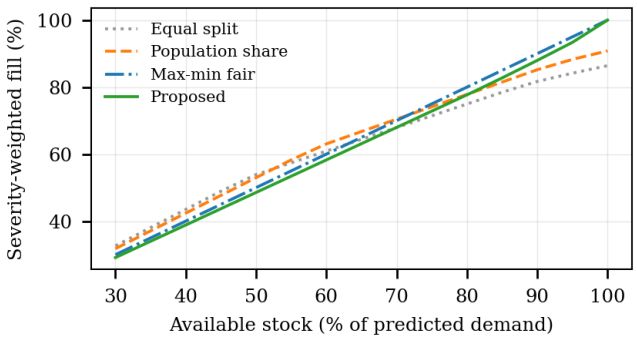


Fig. 7. Severity-weighted fill rate Φ versus available stock.

E. Dynamic Behaviour

Fig. 8 simulates a 48-hour rainfall event in which zones peak at different times. Smoothed severity (Fig. 8a) rises first in Medavakkam, which has the highest vulnerability, and the other zones follow. The priority shares of Eq. (15) (Fig. 8b) shift as zones peak, so the recommended split changes during the event instead of staying fixed. This is the behaviour a static plan cannot provide.

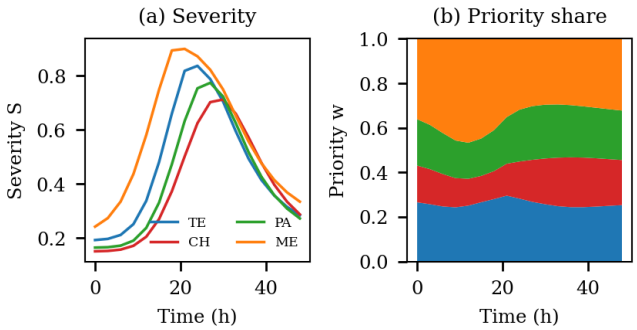


Fig. 8. Simulated 48-hour event: (a) zone severity over time; (b) resulting priority shares.

VIII. DISCUSSION

A. What the Results Mean for Practice

Three observations follow from the experiments. First, a learned demand estimate matters: with severity-dependent, non-linear demand, linear regression leaves an error of about 20% of mean demand, while tree ensembles roughly halve it, and any allocation rule inherits the quality of its demand input. Second, the benefit of need-based allocation depends on how scarce stock is: when stock is far below demand every rule leaves most needs unmet, and when stock is close to demand redistribution decides whether any zone is left short. Third, the choice of priority rule encodes a value judgement. Max-min fairness equalises coverage, population share follows size, and severity priority favours the worst-hit zones; Table IV lets officials see what each choice costs.

B. Deployment Considerations

A deployable system needs reliable feeds of rainfall and water-level data from the responsible agencies, a maintained inventory of depot stock, and current road status. The recommendations should be advisory: officials remain the decision-makers, can inspect the inputs behind each number, and can override a plan. Data on affected populations may be sensitive, so access should be role-based and logged. Because predictions are only as good as the history they are trained on, the models should be re-trained after each real event, and their errors reviewed.

C. Threats to Validity

Internal validity is limited by the synthetic data, which share structure with the models. External validity is limited to four zones and one scenario. Construct validity depends on whether fill rate and the Gini coefficient capture what officials care about, for example time to first delivery or vulnerable-group coverage, which are not modelled here. These threats are the main reason the next step is a study on real event data.

IX. LIMITATIONS

The quantitative results are from a synthetic scenario with assumed parameters; they show how the method behaves, not how it will perform in a real flood. The demand data were generated from the same structure the models learn, which flatters accuracy. Real consumption records are noisy and incomplete. Severity weights are policy parameters rather than learned values. The scenario does not model vehicle capacity, road closures, or delivery delays, and the allocation rule does not optimise Eq. (3) exactly. As Section VII-C shows, the severity-priority rule does not maximise the severity-weighted mean fill under scarcity, and choosing the right priority rule requires input from disaster-management officials. Field validation has not yet been carried out.

X. CONCLUSION AND FUTURE WORK

We presented an integrated, AI-driven platform that links GIS monitoring, environmental indicators, machine-learning demand prediction, and severity-priority dynamic allocation for flood response, with a prototype for four zones around Tambaram, Chennai. In a synthetic evaluation, tree-ensemble models predicted demand with R^2 above 0.97, and under 60% stock the proposed rule gave the most severe zone a markedly higher fill rate than equal and population-share rules while using all available stock, at the cost of a slightly lower severity-weighted mean fill.

Future work includes training on real historical flood and relief data, integrating live rainfall and gauge feeds, extending coverage to more zones, adding vehicle-capacity routing and road-closure updates, solving the linear program of Eq. (3) exactly and comparing it with the closed-form rule, supporting multiple hazards, and piloting the dashboard with local disaster-management officials.

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